Department of Information Technology

Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT
Course Code & Title: CS3391 & Object Oriented Programming
Name of the Faculty member (s): Dr. K. Palraj, ASP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Menu and Menu Items
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO17
- **Activity Chosen:** One Minute Paper

- **Justification:**
  The one-minute paper is used for this topic to quickly assess students' understanding of creating graphical user interfaces using JavaFX. Menus are essential components in desktop applications, and concepts like MenuBar, Menu, and MenuItem help students learn event handling, component hierarchy, and user-interaction design. A short reflective response enables learners to summarize how menus are created, structured, and linked with action events, revealing their grasp of practical implementation.
- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**
  - Faculty explained the concept of creating a MenuBar, Menu, and MenuItem in JavaFX and demonstrated how to link them with action listeners.
  - At the end of the class, the students wrote a one-minute paper summarizing what they had understood.
  - They highlighted how menus are structured and how user actions are handled.
  - Students also mentioned the areas where they needed more clarity.
  - This activity helped assess their understanding and improved their involvement in the class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	3	3	3	3	3	3	3	3	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO9**	**P10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	3	3	3	3	3	3	3	3	3	2
Justification for correlation	Students will be able to applies knowledge of JavaFX components, layouts, and event-handling mechanisms to build functional user interfaces	Students will be able to analyzes GUI requirements and determines suitable JavaFX controls and event-handling logic to address user interactions	Students will be able to designs interactive and user-friendly GUI applications using JavaFX and proper event-handling strategies	Students will be able to use different GUI layouts, control behaviors, and event-handling methods to improve performance	Students will be able to practice Java IDEs and GUI designers to build and debug event-driven JavaFX applications efficiently	Students will be able to works individually or in teams to develop GUI modules, integrate events, and manage application flow	Students will be able to explains the flow of event handling clearly in presentations or code comment	Students will be able to understands the scope, timeline, and resource allocation when writing GUI-based programs	Students are encouraged to learn evolving UI frameworks, styling methods (CSS), and libraries beyond JavaFX	Students will be able to designs user interfaces that support interaction with computational or data-driven applications s	Students will be able to development GUI skills in the areas like mobile apps, IoT dashboards, and smart devices.	Students are encourages modular and maintainable GUI development, following software design practices and UI principles

- **Images / Screenshot of the practice:**

28.10.25

A. Varuna Shree
953624205057

1 Menu and Menu Items.

A menu provides an organized way for users to access commands in an application. It is placed inside a menu bar.

Components used:

Menu Bar - Represents the top-level menu bar in the application window

Menu - Represents a drop-down list of commands

MenuItem - Represents a individual command or option under a menu.

Example:

```
import javafx.application.Application;
import javafx.scene.Scene;
import scene.control.*;
import javafx.scene.layout.BorderPane;
import javafx.stage.Stage
```

These are the required import statements.

Syntax:

1. Creating a Menu Bar - "MenuBar m = new MenuBar()";
2. Creating a Menu - "Menu a = new Menu("MenuName");
   Menu filemenu = new Menu("File")
3. Creating Menu Items - MenuItem itemName = new MenuItem ("ItemName")
4. Adding Menu Item to a menu - menuName.getItems().addAll(item1, 2, 3);

Menu items can perform specific actions when clicked. The method setOnAction() is used to handle these events.

Menus are usually added to layout such as Border Pane by placing the menubar at the top using root.setTop (menuBar)

In conclusion, menu and menuItems in JavaFx help create well structured and user friends interfaces.

Figure 1. Sample One-Minute Paper of A. Varuna Sree

903624205048.
Santhiya S.

Menu and Menu Item.

*A menu is a list of options or commands in a software application that helps users perform task easily. It improves user interaction and navigation in graphical user interfaces.

Types of Menus:-

1. MenuBar - Display main options like File, Edit, View.
2. Drop-Down Menu - Appears when a menu is clicked
3. Context menu - opens with a right - click.
4. Submenu - A menu inside another menu.

*A menu item is a individual option with in a menu such as new, open, save or exit. Selecting a menuItem perform a specific action.

Example. Program:-

```
public void start(Stage primary Stage) throws Exception
{
 menu newmenu = new menu("File")
 menu newmenu = new menu("Edit")
 menuItem m1 = new menuItem("open")
 menuItem m2 = new menuItem("save")
 menuItem m5 = new menuItem("copy")
 menuItem m6 = new menuItem("paste");
 newmenu.getItems.add(m1)
 newmenu.getItems.add(m2)
 newmenu.getItems.add(m5)
 newmenu.getItems.add(m6)
 menuBar newmb = new menuBar();
 newmb.getmenus().add(newmenu);
 newmb.getmenus().add(newmenu2);
 VBox box = new VBox(newmb);
 Scene scene = new Scene (box, 400, 300);
 PrimaryStage.setScene(Scene)
 PrimaryStage.setTitle("JavaFX Menu Example);
} Primary Stage.Show();
public static void main (String[] args)
}
```

Figure 2 Sample One-Minute Paper of S.Santhiya

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
    - Students expressed that the practice session helped them clearly understand how Java Menu, MenuBar, and MenuItem work in real applications.
    - Many appreciated the hands-on demonstration, stating it improved their confidence in implementing GUI components.
    - Some students requested additional examples on event handling to strengthen their coding skills.
- ***Benefit of the practice:***
    - Students were actively participated in this activity.
    - Students can able to write the topic without any confusion.
    - It improves confidence in designing Java GUI applications to the students.
    - Make the students to know the impact and importance of writing their views and understanding related to the topic and made them involve in the activity.
- ***Challenges faced in implementation:***
    - Some students found it difficult to understand the hierarchy of Menu, MenuItem.
    - Some students are unable to handling multiple action events using listeners caused confusion.
    - Some students are limited experience with GUI development made it harder to visualize the final output while coding.

References:

- https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html
- https://www.unl.edu/gradstudies/current/teaching/minute
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/GM_CS3491_One_Minute_Paper.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/KP_CS3591_Error_Control.pdf /

Signature of Faculty Member **HOD**